

1. (Section 7.1, Problem 3) Test the graph of  $y = x^3 - x^2$  for symmetry about the  $x$ -axis,  $y$ -axis and the origin.

$$\begin{array}{lll}
 \text{x-axis} & y \rightarrow -y & y = x^3 - x^2 \rightarrow -y = x^3 - x^2 \quad \times \\
 \text{y-axis} & x \rightarrow -x & y = x^3 - x^2 \rightarrow y = (-x)^3 - (-x)^2 \\
 & & y = -x^3 - x^2 \quad \times \\
 \text{origin} & \begin{cases} x \rightarrow -x \\ y \rightarrow -y \end{cases} & y = x^3 - x^2 \rightarrow -y = (-x)^3 - (-x)^2 \\
 & & -y = -x^3 - x^2 \\
 & & y = x^3 + x^2 \quad \times
 \end{array}$$

No symmetry

2. (Section 7.1, Problem 9) Test the graph of  $y = \frac{x}{x^2 + 1}$  for symmetry about the  $x$ -axis,  $y$ -axis and the origin.

$$\begin{array}{lll}
 \text{x-axis} & y \rightarrow -y & y = \frac{x}{x^2 + 1} \rightarrow -y = \frac{x}{x^2 + 1} \quad \times \\
 \text{y-axis} & x \rightarrow -x & y = \frac{x}{x^2 + 1} \rightarrow y = \frac{-x}{(-x)^2 + 1} \\
 & & y = \frac{-x}{x^2 + 1} \quad \times \\
 \text{origin} & \begin{cases} x \rightarrow -x \\ y \rightarrow -y \end{cases} & y = \frac{x}{x^2 + 1} \rightarrow -y = \frac{-x}{(-x)^2 + 1} \\
 & & -y = \frac{-x}{x^2 + 1} \\
 & & y = \frac{x}{x^2 + 1} \quad \checkmark
 \end{array}$$

Origin symmetry

3. (Section 7.1, Problem 19) Find all of the intercepts of the graph of

$$y = \frac{(2x+1)(x-1)^2}{(x-2)(x+1)^2}$$

x-axis  $(2x+1)(x-1)^2 = 0 \Rightarrow 2x+1=0 \text{ or } x-1=0$   
 $x = -\frac{1}{2} \quad x=1$

$$\left(-\frac{1}{2}, 0\right), (1, 0)$$

y-axis  $y = \frac{(2x+1)(x-1)^2}{(x-2)(x+1)^2} \Big|_{x=0} = \frac{(1)(-1)^2}{(-2)(1)^2} = -\frac{1}{2}$

$$\left(0, -\frac{1}{2}\right)$$